

(1001CJA621121001)

Test Pattern

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2021 - 2022)

JEE(Advanced)
UNIT TEST
14-02-2021
JEE(Main + Advanced) : NURTURE COURSE (PHASE : TEPS)**ANSWER KEY****PAPER-1****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,C,D	A,B,C	A,B	A,B,C	A,C	A,C	A	A	B	A
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	0.41 to 0.42	85.50	1.25	1.33	5.00	0.56	80.00	2.00		

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,B	B,D	A,B,D	A,D	A,B,C,D	A,B	A	C	D	A
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	8.30 to 9.30	7.50	1.75	1477.80 to 1478.00	0.00	8.00	4.50	1.25		

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,D	B,C	A,B,D	C	B,D	A,B,C	B	A	B	B
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	5.00	72.00	3456.00	9.00	4.00	5.00	3.00	7.00		

(1001CJA621121002)

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(Academic Session : 2021 - 2022)

JEE(Main)
UNIT TEST
14-02-2021
JEE(Main + Advanced) : NURTURE COURSE (PHASE : TEPS)**ANSWER KEY****PAPER-2****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	A	D	B	C	A	A	D	D	C
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	C	C	C	D	A	B	C	C	B	C
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	41.66 to 41.67	8.00	4.00	48.00	2.00	5.00	4.00	5.00	6.00	3.00

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	C	A	A	A	D	B	A	B	A
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	B	A	D	B	B	B	B	D	A
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	4.00	9.00	4.00	3.00	3.25	3.00	4.00	25.00	80.85	0.47

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	A	A	B	B	D	D	C	C	C
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	C	D	A	B	D	B	C	C	B
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	3.00	20.00	-4.00	8.00	-1.00	108.00	12.00	10.00	3.00	589.00



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(Academic Session : 2021 - 2022)

JEE(Advanced)

UNIT TEST

14-02-2021

JEE(Main + Advanced) : NURTURE COURSE (PHASE : TEPs)

PAPER-1

PART-2 : CHEMISTRY SOLUTION

SECTION-I

1. Ans.(A, B)

2. Ans.(B, D)

3. Ans.(A,B,D)

Initial condition : $PV = nRT$

$$\text{or, } 1.05 \times 2V = 0.7 \times 0.083 \times 300 \Rightarrow V = 8.3 \text{ L}$$

$$\text{Final condition : } n_1 \times R \times 500 = n_2 \times R \times 200$$

$$\text{and } n_1 + n_2 = 0.7$$

$$\text{Hence, } n_1 = 0.2, n_2 = 0.5$$

$$\text{and } P = \frac{n_1 \times R \times 500}{V} = 1.0 \text{ bar}$$

4. Ans.(A,D)

5. Ans.(A,B,C,D)

6. Ans.(A, B)

7. Ans.(A)

8. Ans.(C)

9. Ans.(D)

$$n_{\text{Cl}_2} = 0.1$$

$$n_{\text{Cl}_2 \text{ withdraw}} = \frac{1 \times 246 \times 10^{-3}}{0.082 \times 300} = 0.01$$

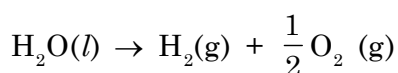
$$n_{\text{Cl}_2 \text{ left}} = 0.09$$

10. Ans.(A)

SECTION-II

1. Ans. (8.30 to 9.30)

$$P_{\text{N}_2} = 1 - 0.04 = 0.96 \text{ atm}$$



$$\begin{array}{cccc} \text{Initial} & x \text{ mole} & 0 & 0 \\ \text{Final} & 0 & x & x/2 \end{array}$$

$$\text{Now, } \frac{3x}{2} = \frac{(1.86 - 0.96 - 0.04) \times 20}{0.08 \times 300}$$

$$\Rightarrow x = \frac{4.3}{9}$$

$$\therefore \text{Mass of liquid water present initially} \\ = x \times 18 = 8.6 \text{ gm}$$

2. Ans.(7.50)

3. Ans.(1.75)

$$\text{Fe}_2(\text{SO}_4)_3 = x \text{ moles}$$

$$\text{FeC}_2\text{O}_4 = y \text{ moles}$$

$$y \times 3 = 40 \times \frac{1}{16}$$

$$2x + y = 60 \times \frac{1}{16}$$

4. Ans.(1477.80 to 1478.00)

5. Ans.(0.00)

6. Ans.(8.00)

7. Ans.(4.50)

8. Ans.(1.25)

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. **Ans. (A,D)**

Sol. $D = (\alpha + \beta - \gamma)(\beta + \gamma - \alpha)(\gamma + \alpha - \beta)$

$$D = (\alpha - \gamma)(\beta - \alpha)(\gamma - \beta)$$

$$D = -(\alpha - \beta)(\beta - \gamma)(\gamma - \alpha)$$

$$D_{\min} = -(3-2)(2-1)(1-3) = 2$$

for any combination of α, β & γ , D is always an even integer.

2. **Ans. (B,C)**

Sol. $\cos A + 2\cos B + \cos C = 2$

$$\Rightarrow \cos A + \cos C = 2(1 - \cos B)$$

$$\Rightarrow 2\cos \frac{A+C}{2} \cos \frac{A-C}{2} = 2 \left(2\sin^2 \frac{B}{2} \right)$$

$$\Rightarrow \sin \frac{B}{2} \cos \frac{A-C}{2} = 2\sin^2 \frac{B}{2} \quad \left(\because \frac{A+C}{2} = \frac{\pi}{2} - \frac{B}{2} \right)$$

$$\Rightarrow \cos \left(\frac{A-C}{2} \right) = 2\cos \left(\frac{A+C}{2} \right)$$

$$\Rightarrow \cos \frac{A}{2} \cos \frac{C}{2} + \sin \frac{A}{2} \sin \frac{C}{2}$$

$$= 2\cos \frac{A}{2} \cos \frac{C}{2} - 2\sin \frac{A}{2} \sin \frac{C}{2}$$

$$\Rightarrow 3\sin \frac{A}{2} \sin \frac{C}{2} = \cos \frac{A}{2} \cos \frac{C}{2}$$

$$\Rightarrow \tan \frac{A}{2} \tan \frac{C}{2} = \frac{1}{3} \Rightarrow \cot \frac{A}{2} \cot \frac{C}{2} = 3 \quad \dots (i)$$

Hence (B)

In a triangle

$$\cot \frac{A}{2} + \cot \frac{B}{2} + \cot \frac{C}{2} = \cot \frac{A}{2} \cot \frac{B}{2} \cot \frac{C}{2}$$

$$\Rightarrow \cot \frac{A}{2} + \cot \frac{B}{2} + \cot \frac{C}{2} = 3\cot \frac{B}{2} \quad (\because \text{by (i)})$$

$$\Rightarrow \cot \frac{A}{2} + \cot \frac{C}{2} = 2\cot \frac{B}{2} \quad \text{Hence (C)}$$

3. **Ans. (A,B,D)**

Sol. $\sin A : \sin B : \sin C \Rightarrow 4 : 7 : 9$

from sine rule $a : b : c \Rightarrow 4 : 7 : 9$

$$\text{Area of } \triangle BIC = \frac{1}{2} ar, \text{ Area of } \triangle CIA = \frac{1}{2} br$$

$$\text{then ratio of areas} = \frac{1}{2} ar : \frac{1}{2} br : \frac{1}{2} cr = a : b : c$$

we know that

area of $\triangle BGC$ = area of $\triangle CGA$ = area of $\triangle AGB$

so ratio of areas = 1 : 1 : 1

length of tangents are $s-a$, $s-b$, $s-c$

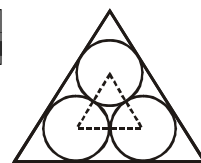
so their ratios are $(10-4) : (10-7) : (10-9)$

4. **Ans. (C)**

$$\text{Sol.} \left(\frac{\sqrt{3}}{4} \cdot 16 - 3 \cdot \frac{1}{2} \cdot 4 \cdot \frac{\pi}{3} \right) + 3 \left[2 \cdot 4 - \frac{2\pi \cdot 4}{4} \right]$$

$$= 4\sqrt{3} - 2\pi + 24 - 6\pi$$

$$= 24 + 4\sqrt{3} - 8\pi$$



5. **Ans. (B,D)**

$$\text{Sol.} D = \begin{vmatrix} \lambda+1 & 1 & \lambda \\ \lambda & 2\lambda & 1 \\ 3 & \lambda-1 & \lambda \end{vmatrix}$$

$$= 3\lambda^3 - 7\lambda^2 + 4 = (\lambda-1)(\lambda-2)(3\lambda+2) \neq 0$$

for unique solution

$$\Rightarrow \lambda \in \mathbb{R} - \{-2/3, 1, 2\} \text{ for unique solution}$$

6. **Ans. (A,B,C)**

$$\text{Sol.} (a+2)x^2 + bx + c = 0 \quad \dots (i)$$

$$2x^2 + 3x + 4 = 0 \quad \dots (ii)$$

for (ii) $D < 0$

$\Rightarrow$ (i) & (ii) have both roots common.

$$\therefore \frac{a+2}{2} = \frac{b}{3} = \frac{c}{4}$$

$\therefore$ for equation (1) $D < 0$

$$\Rightarrow b^2 - 4c(a+2) < 0$$

$$b^2 < 4ac + 8c.$$

$$\frac{a+2}{2} = \frac{b}{3} = \frac{c}{4} = k(\text{say}) \Rightarrow \{k \in \mathbb{N}, k \geq 2\}$$

$$\Rightarrow b^2 - 4ac = k(32 - 23k) < 0$$

$$\text{Also, } \frac{a+b+c+2}{9} = k \Rightarrow a+b+c = 9k-2$$

$$(a+b+c)_{\min} = 16$$

Paragraph for Question 7 & 8

7. **Ans. (B)**

Sol. From (i) $a_{11} = 2, a_{12} = 3 = a_{21}, a_{31} = 5 = a_{22} = a_{13}$
 $a_{32} = 7 = a_{23}, a_{33} = 11$

$$\Rightarrow D_1 = \begin{vmatrix} 2 & 3 & 5 \\ 3 & 5 & 7 \\ 5 & 7 & 11 \end{vmatrix} = 12 + 6 + (-20) = -2$$

$$\Rightarrow S_1 = 48 \text{ and } \Delta_1 = -2 \Rightarrow S_1 + \Delta_1 = 46$$

8. **Ans. (A)**

Sol. $\therefore$ least entry of D_2 is 1 $\Rightarrow a_{11} = 1$
 and for minimum value of $S_2, a_{12} = 4 = a_{21}$
 $a_{31} = 6 = a_{22} = a_{13}, a_{32} = 8 = a_{23}, a_{33} = 9$

$$\Rightarrow D_2 = \begin{vmatrix} 1 & 4 & 6 \\ 4 & 6 & 8 \\ 6 & 8 & 9 \end{vmatrix} = -10 + 48 - 24 = 14$$

$$\Rightarrow \text{Minimum value of } S_2 = 52$$

Paragraph for Question 9 & 10

$$a + b + c = 5 \text{ (a, b, c > 0)} \quad x^2 y^3 = 243 = 3^5$$

9. **Ans. (B)**

Sol. Using $AM \geq GM$

$$\frac{x^2 + y + y + y + 1}{5} \geq (x^2 \cdot y \cdot y \cdot y \cdot 1)^{1/5}$$

$$\Rightarrow x^2 + 3y + 1 \geq 5 \cdot (243)^{1/5} \geq 15$$

$$\text{But } x^2, y \neq 1, \text{ hence } x^2 + 3y + 1 > 15$$

10. **Ans. (B)**

$$\text{Sol. } \frac{x + x + y + y + y}{5} \geq (x^2 y^3)^{1/5} \geq \frac{5}{\frac{2}{x} + \frac{3}{y}}$$

$$\Rightarrow \frac{2x + 3y}{5} \geq 3 \geq \frac{5xy}{3x + 2y}$$

SECTION-II

1. **Ans. 5.00**

Sol. Number is divisible by 9 and 8

The Number is divisible by 8 if last three digits are divisible by 8 i.e. 79b is divisible by 8 $\Rightarrow b = 2$

For divisible by 9 $a + 6 + 7 + 9 + b$

is divisible by 9 $\Rightarrow a + 24$ is divisible by 9

$$\Rightarrow a = 3$$

$$\Rightarrow a + b = 5$$

2. **Ans. 72.00**

Sol. Draw parallelogram OBDC.

OE = ED & as

$$OC^2 = OD^2 + DC^2,$$

ΔODC is a

right angled $\Delta(6, 8, 10)$

$\Rightarrow$ Area of ΔADC

$$= \frac{1}{2} \times 8 \times 12 = 48$$

$$\Delta EDC = \frac{1}{2} \times 3 \times 8 = 12$$

$$\text{Area } \Delta AEC = \text{Area } \Delta ADC - \text{Area } \Delta EDC$$

$$= 48 - 12 = 36$$

$$\text{Area } \Delta AEC = 36 \Rightarrow \text{Area } \Delta ABC = 72$$

3. **Ans. 3456.00**

Sol. Let T_n be first term of n^{th} row

let

$$S_n = 1 + 3 + 7 + 13 + 21 + \dots + T_n$$

$$S_n = 1 + 3 + 7 + 13 + \dots + T_n$$

$$T_n = 1 + 2 + 4 + 6 + \dots \text{upto to } n \text{ terms}$$

$$T_n = 1 + n(n-1) = n^2 - n + 1$$

$$T_{12} = 133$$

sum of the numbers in horizontal

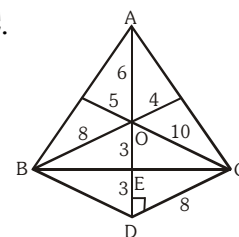
diagonal = $133 + 135 + \dots \text{upto } 12 \text{ terms}$

$$= \frac{12}{2} [2 \times 133 + 11 \times 2]$$

$$= 12 \times 144 = 1728$$

then sum of the numbers in vertical and

horizontal diagonal = $2 \times 1728 = 3456$.



4. Ans. 9.00

Sol. $n(P) = 38$ $n(P \cup A \cup I) = 58$

$n(A) = 20$ $n(P \cap A \cap I) = 3$

$n(I) = 15$

Now

$$n(P \cup A \cup I) = n(P) + n(A) + n(I) - [n(P \cap A) + n(A \cap I) + n(I \cap P)] + n(P \cap A \cap I)$$

$$\Rightarrow n(P \cap A) + n(A \cap I) + n(I \cap P)$$

$$= 38 + 15 + 20 - 58 + 3 = 18$$

Now number of students who got selected in exactly two exams

$$= n(P \cap A) + n(A \cap I) + n(I \cap P) - 3n(P \cap A \cap I)$$

$$= 18 - 3 \times 3 = 9$$

OR Solve by venn diagram.

5. Ans. 4.00

Sol. $\left. \begin{array}{l} x_1 x_2 = 2 \\ x_3 x_4 = 4 \\ x_5 x_6 = 6 \\ x_7 x_8 = 8 \end{array} \right\}$ therefore, $x_1 x_2 \dots x_8 = 2.4.6.8 =$

$$384$$

$$\log_{3\sqrt{2}}(384 - 60) = \log_{3\sqrt{2}} 324 = 4$$

6. Ans. 5.00

Sol. $\alpha + \gamma = \frac{4}{A}$ and $\beta + \delta = \frac{6}{B}$

$$\alpha\gamma = \frac{1}{A} \quad \beta\delta = \frac{1}{B}$$

As $\alpha, \beta, \gamma, \delta$ are in H.P.

$$\beta = \frac{2\alpha\gamma}{\alpha + \gamma} = \frac{1}{2} \quad \& \quad \gamma = \frac{2\beta\delta}{\beta + \delta} = \frac{1}{3}$$

Now $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}, \frac{1}{\delta}$ are in A.P.

$$\Rightarrow \frac{1}{\alpha}, 2, 3, \frac{1}{\delta} \text{ are in A.P.}$$

$$\Rightarrow \frac{1}{\alpha} = 1 \quad \& \quad \frac{1}{\delta} = 4$$

$$\Rightarrow \alpha = 1 \quad \text{and} \quad \delta = \frac{1}{4}$$

$$\Rightarrow A = \frac{1}{\alpha\gamma} = 3 \quad \& \quad B = \frac{1}{\beta\delta} = 8$$

$$\Rightarrow B - A = 5$$

7. Ans. 3.00

Sol. $2^{(\sin x + \cos x)\sqrt{1+2y-y^2}} \geq 2^{\frac{y^2+1}{y^2}}$

we know that

$$\sin x + \cos x \leq \sqrt{2}$$

$$\sqrt{1+2y-y^2} = \sqrt{-(y-1)^2+2} \leq \sqrt{2}$$

$$\Rightarrow \text{L.H.S.} \leq 2^2 = 4$$

$$\text{but R.H.S.} \geq 2^2 = 4$$

solution can be obtained if L.H.S. = R.H.S. = 4

$$\Rightarrow x = -2\pi + \frac{\pi}{4}, \frac{\pi}{4}, 2\pi + \frac{\pi}{4} \text{ and } y = 1.$$

8. Ans. 7.00

Sol. Let $|x^2 - 4x| = a$ & $|x^2 - 2x| = b$

$$\Rightarrow a^b > 1 \quad \& \quad b^a > 1$$

$$\Rightarrow \text{if } a < a < 1, \text{ then } b < 0 \text{ (not possible)}$$

$$\therefore a > 1 \quad \Rightarrow \quad b > 0$$

$$\text{Similarly } b > 1 \quad \Rightarrow \quad b > 0 \text{ for } b^a > 1$$

$$\therefore a > 1 \quad \& \quad b > 1$$

$$\Rightarrow |x^2 - 4x| > 1$$

$$\Rightarrow x^2 - 4x - 1 > 0 \quad \text{or} \quad x^2 - 4x + 1 < 0$$

$$\Rightarrow x > 2 + \sqrt{5} \quad \text{or} \quad 2 - \sqrt{3} < x < 2 + \sqrt{3} \quad \dots(i)$$

$$|x^2 - 2x| > 1$$

$$\Rightarrow x^2 - 2x > 1 \quad \text{or} \quad x^2 - 2x < -1$$

$$\Rightarrow x > 1 + \sqrt{2} \quad \dots(ii)$$

$$\Rightarrow (x-1)^2 < 0 \quad \text{or} \quad x < 1 - \sqrt{2}$$

$$\Rightarrow (x-1)^2 < 0, x \in \phi$$

$$\text{or } x < 1 - \sqrt{2}$$

combining (i) and (ii) we get

$$x \in (-\infty, 1 - \sqrt{2}) \cup (1 + \sqrt{2}, 2 + \sqrt{3})$$



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PAPER-2

PART-2 : CHEMISTRY SOLUTION

SECTION-I

1. Ans. (C)

2. Ans. (C)

$$U_{se} = PV = nRT$$

$$P_1 = 5 \quad V = V \quad T = 273 \text{ K}$$

$$P_2 = ? \quad V = 1/3V \quad T = 829 \text{ K}$$

3. Ans. (A)

$$P_T = P_{O_2} + P_{H_2O}$$

$$720 = P_{O_2} + 20 \Rightarrow P_{O_2} = 700$$

$$V \rightarrow 4V$$

$$P \rightarrow \frac{1}{4} = \frac{700}{4} = \frac{350}{2} = 175$$

$$\Rightarrow 175 \text{ mm of Hg}$$

4. Ans. (A)

5. Ans. (A)

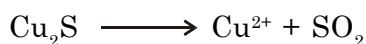
$$\text{Normality of NaOH} = x$$

$$\text{Molarity} = x$$

$$N_{Ba(OH)_2} = Y, M = \frac{Y}{2}$$

$$V_1 \times X + V_2 \times Y = 100 \times 0.1$$

6. Ans. (D)



$$n_b = 8$$



$$n_b = 6$$

7. Ans. (B)



$$1 \quad 2 \quad 0 \quad 0$$

$$\Rightarrow 0 \quad 1 \quad 1 \quad 1$$

$$\text{So 1 mole } H_2S_4 \text{ \& 1 mole } H_2F_2$$

$$= 4 \text{ mole } HO^- = 2 \text{ mole } Ca(OH)_2$$

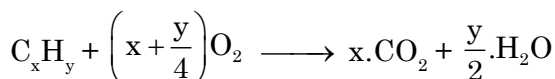
8. Ans. (A)

9. Ans. (B)



$$Cr \text{ will be } +5$$

10. Ans. (A)



$$10 \text{ ml}$$

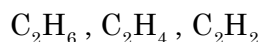
$$10x$$

$$\text{volume contraction} = \text{volume by KOH absorbed} = \text{volume of } CO_2$$

$$20 \text{ ml} = x \times 10 \text{ ml}$$

$$x = 2$$

$$\text{so hydrocarbon may be}$$



11. Ans. (A)

12. Ans. (B)

13. Ans. (A)

14. Ans. (D)

15. Ans. (B)

16. Ans. (B)

17. Ans. (B)

18. Ans. (B)

19. Ans. (D)

20. Ans. (A)

SECTION-II

1. Ans. (4.00)

2. Ans. (9.00)

3. Ans. (4.00)

4. Ans. (3.00)

5. Ans. (3.25)

6. Ans. (3.00)

 Use $PV = nRT$

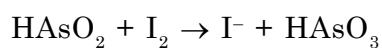
$$T_b = 2T_i \quad P_b = \frac{P_i}{2}$$

 Find n_{final} as $V = \text{constant}$

7. Ans. (4.00)

$$\text{Use } P = \frac{dRT}{M}$$

8. Ans. (25.00)



$$\text{Gm eq. of HAsO}_2 = \text{Gm eq HAsO}_3$$

9. Ans. (80.85)

$$n_{\text{FeO}} = a \quad n_{\text{Fe}_2\text{O}_3} = b$$

$$a \times 1 = 100 \times 0.25 \times 5$$

$$a \times 1 + b \times 2 = 1000 \times 0.1 \times 6$$

10. Ans. (0.47)

$$n_{\text{NO}_3^-} = n_{\text{NH}_3} \text{ as nitrogen conserved}$$

$$\text{so, } 25 \times M \times 1 + 32 \times 0.1 = 100 \times 0.15$$

$$(\text{NH}_3) \quad (\text{NaOH}) \quad (\text{HCl})$$

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. Ans. (A)

$$\Delta = (xyz)^n \begin{vmatrix} 1 & x^2 & x^4 \\ 1 & y^2 & y^4 \\ 1 & z^2 & z^4 \end{vmatrix} = (xyz)^n (x^2 - y^2)(y^2 - z^2)(z^2 - x^2)$$

Clearly when $n = -4$,

$$\Delta = \left(\frac{1}{\mathbf{v}^2} - \frac{1}{\mathbf{x}^2} \right) \left(\frac{1}{\mathbf{z}^2} - \frac{1}{\mathbf{v}^2} \right) \left(\frac{1}{\mathbf{x}^2} - \frac{1}{\mathbf{z}^2} \right)$$

2. Ans. (A)

$$195 \{(3x^3)^3-1\} + 3(1 + 3x^3 + 9x^6) = 0$$

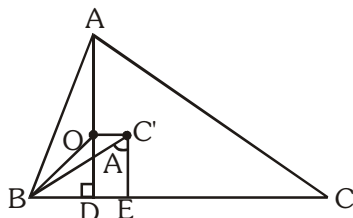
$$195\{(3x^3)-1\} + 3\left\{\frac{1-(3x^3)^3}{1-3x^3}\right\} = 0$$

$$195 \{(3x^3)^3 - 1\}\{3x^3 - 1\} + 3\{(3x^3)^3 - 1\} = 0$$

$$((3x^3)^3 - 1) \{195(3x^3 - 1) + 3\} = 0$$

No natural solution.

3. Ans. (A)



According to question.

$$OD = C'E$$

$$\Rightarrow 2R \cos B \cos C = R \cos A$$

$$\Rightarrow 2 \cos B \cos C = -\cos (B + C)$$

$$(\because \cos A = \cos (\pi - (B + C)))$$

$$\Rightarrow 2 = \frac{-\cos(B+C)}{\cos B \cdot \cos C} \Rightarrow \tan B \cdot \tan C = 3.$$

4. **Ans. (B)**

$\therefore a_{12} = 1, a_{22} = 2$ and $a_{32} = 1$ are the elements of second column

$\Rightarrow$ their cofactors become $-a$, b and $-c$ respectively

$$\Rightarrow -a + 2b - c = D$$

$$\Rightarrow a - 2b + c = -D$$

5. Ans. (B)

$$2\left(\cos\left(\pi\cos^2 x\right)\right)^2 = 1 + \cos\left(\frac{\pi\cos 2x}{2}\right)$$

$$2\left(\cos\left(\frac{\pi}{2} + \frac{\pi \cos 2x}{2}\right)\right)^2 = 1 + \cos\left(\frac{\pi \cos 2x}{2}\right)$$

$$2\sin^2\left(\frac{\pi\cos 2x}{2}\right)=1+\cos\left(\frac{\pi\cos 2x}{2}\right)$$

Let $\frac{\pi \cos 2x}{2} = t$

$$2(1 - \cos^2 t) = 1 + \cos t$$

$$2\cos^2 t + \cos t - 1 = 0$$

$$(2\cos t - 1)(\cos t + 1) = 0$$

$$\text{cost} = \frac{1}{2} \quad \& \quad \text{cost} = -1$$

$$\cos\left(\frac{\pi}{2}\cos 2x\right)=\frac{1}{2}$$

$$\Rightarrow \cos\left(\frac{\pi}{2} - \pi \sin^2 x\right) = \frac{1}{2}$$

$$\sin(\pi \sin^2 x) = \frac{1}{2}$$

6. Ans. (D)

Let $R = 0.ab\ddot{c}d$

Thus, we have $100R = ab.\dot{c}\dot{d}$

& 10000R = abcd. $\overset{\cdot\cdot}{c}\overset{\cdot\cdot}{d}$

Subtracting we have, $9900R = abcd - ab$

$$= (1000a + 100b + 10c + d) - (10a + b)$$

$$= 990a + 99b + 10c + d$$

$$\text{i.e. } R = \frac{a}{10} + \frac{b}{100} + \frac{c}{990} + \frac{d}{9900}$$

7. Ans. (D)

$$\begin{vmatrix} x_1 & x_2 & x_3 \\ x_4 & x_5 & x_6 \\ x_7 & x_8 & x_9 \end{vmatrix}$$

$$100 C_1 + 10 C_2 + C_3 \rightarrow C_3$$

$$= \begin{vmatrix} x_1 & x_2 & x_1 x_2 x_3 \\ x_4 & x_5 & x_4 x_5 x_6 \\ x_7 & x_8 & x_7 x_8 x_9 \end{vmatrix}$$

$$\text{Let } x_1 x_2 x_3 = 8k, x_4 x_5 x_6 = 12\ell, x_7 x_8 x_9 = 16m$$

$$\begin{vmatrix} x_1 & x_2 & 8k \\ x_4 & x_5 & 12\ell \\ x_7 & x_8 & 16m \end{vmatrix} = 4 \begin{vmatrix} x_1 & x_2 & 2k \\ x_4 & x_5 & 3\ell \\ x_7 & x_8 & 4m \end{vmatrix}$$

Hence determinant must be divisible by 4.

8. Ans. (C)

$$2 < x < 3 \Rightarrow x - 3 < 0 \text{ and } x - 2 > 0$$

$$\Rightarrow (x - 3)(x - 2) < 0 \quad \forall \quad x \in (2, 3)$$

9. Ans. (C)

$$\frac{\text{Perimeter}_{DEF}}{\text{Perimeter}_{ABC}} = \frac{R(\sin 2A + \sin 2B + \sin 2C)}{a + b + c}$$

$$= \frac{4 \sin A \cdot \sin B \cdot \sin C}{2(\sin B + \sin A + \sin C)}$$

$$= \frac{4 \times 8(\pi \sin A / 2)(\pi \cos A / 2)}{2 \times 4(\pi \cos A / 2)} = \frac{r}{R}$$

$$\therefore (r = 4R \pi \sin \frac{A}{2})$$

10. Ans. (C)

If we divide $x^3 - k$ by $x^2 + x + p$, then the remainder should be zero

$$\begin{array}{r} x^2 + x + p \overline{) \begin{array}{r} x^3 - k \\ x^3 + x^2 + px \\ \hline -x^2 - px - k \\ -x^2 - x - p \\ \hline (1-p)x + p - k \end{array}} \end{array}$$

$$\Rightarrow p = 1 \quad \& \quad p = k$$

$$\Rightarrow k = 1 \quad \Rightarrow \quad p + k = 2$$

11. Ans. (B)

Let the roots of $x^2 - 2\sqrt{3}x + 2 = 0$ are a & b

$$\Rightarrow a = \sqrt{3} + 1, b = \sqrt{3} - 1$$

$$\text{and included angle } C = \frac{\pi}{3}$$

$$\Rightarrow c = \sqrt{a^2 + b^2 - 2ab \cos C}$$

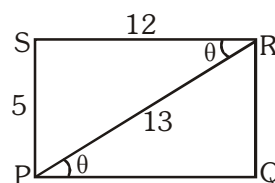
$$= \sqrt{(4 + 2\sqrt{3}) + (4 - 2\sqrt{3}) - 2(2) \cdot \frac{1}{2}} = \sqrt{6}$$

$\therefore$ Perimeter of the triangle

$$= \sqrt{3} + 1 + \sqrt{3} - 1 + \sqrt{6} = 2\sqrt{3} + \sqrt{6}$$

12. Ans. (C)

$$PR + PQ = 5PS$$



$$\Rightarrow \frac{PR}{PS} + \frac{PQ}{PS} = 5$$

$$\Rightarrow \operatorname{cosec} \theta + \cot \theta = 5 \quad \dots\dots(i)$$

$$\operatorname{cosec} \theta - \cot \theta = \frac{1}{5} \quad \dots\dots(ii)$$

$$(i) + (ii) \Rightarrow \operatorname{cosec} \theta = \frac{13}{5}$$

$$\Rightarrow PR = 13 \quad \therefore PR - PS = 8$$

$$PS = 5$$

$$\Rightarrow \text{Area} = 12 \times 5 = 60 \text{ sq. units}$$

13. Ans. (D)

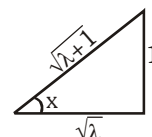
$$\lambda \tan^2 x + \frac{1}{\lambda \tan^2 x} = 2$$

$$\Rightarrow \lambda \tan^2 x = 1 \Rightarrow \tan x = \frac{1}{\sqrt{\lambda}}$$

$$\sin x = \frac{1}{\sqrt{\lambda+1}} \quad \& \quad \cos x = \frac{\sqrt{\lambda}}{\sqrt{\lambda+1}}$$

$$\sin^2 x + 5 \cos^2 x + 3 \sin x \cos x$$

$$= \frac{1}{(\lambda+1)} + \frac{5\lambda}{(\lambda+1)} + \frac{3\sqrt{\lambda}}{(\lambda+1)} = \frac{1+5\lambda+3\sqrt{\lambda}}{(\lambda+1)}$$



14. Ans. (A)

$$\lambda = \frac{t^2 + 4}{t^2 + 3t + 4}$$

λ is always positive

Now $\Delta = \begin{vmatrix} 3 & -1 & 4 \\ 1 & 2 & -3 \\ 6 & 5 & \lambda \end{vmatrix} = 7(\lambda + 5) \neq 0$

since λ is always positive. Hence the given system has unique solution.

15. Ans. (B)

$$\cos A = \frac{-3}{5} \Rightarrow \sin A = \frac{4}{5} \text{ \& \; } \tan A = -\frac{4}{3}$$

$$\therefore \sin A + \tan A = \frac{4}{5} - \frac{4}{3} = \frac{-8}{15} \text{ \& \; } \sin A \cdot \tan A = \frac{-16}{15}$$

$\Rightarrow$ Required quadratic equation is

$$x^2 - \left(\frac{-8}{15}\right)x + \left(\frac{-16}{15}\right) = 0 \text{ or } 15x^2 + 8x - 16 = 0$$

16. Ans. (D)

$\log(a + c), \log(a + b), \log(b + c)$ are in A.P.

$$(a + b)^2 = (a + c)(b + c)$$

$$(a + b)^2 = ab + c(a + b) + c^2 \quad \dots\dots (i)$$

$$a, c, b \text{ are in H.P.} \quad \therefore c = \frac{2ab}{a+b}$$

$$\Rightarrow ab = \frac{c(a+b)}{2} = \frac{kc^2}{8}$$

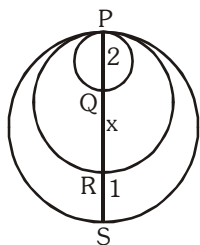
$$\text{from (i), } \frac{k^2c^2}{16} = \frac{3kc^2}{8} + c^2$$

$$\Rightarrow k^2 - 6k - 16 = 0 \Rightarrow k = 8 \text{ or } -2$$

$$\therefore k = 8 \quad \{k > 0\}$$

17. Ans. (B)

Sol.



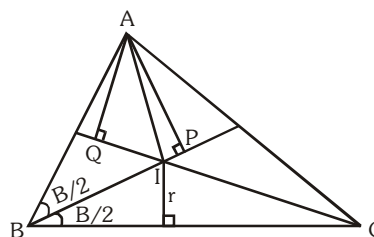
$$\pi r_1^2 + \pi r_3^2 = 2\pi r_2^2$$

$$\Rightarrow 1^2 + \left(\frac{x+3}{2}\right)^2 = 2\left(\frac{x+2}{2}\right)^2$$

$$x = \sqrt{6} - 1$$

18. Ans. (C)

Sol.



In triangle API

$$AP = AB \sin \frac{B}{2}$$

Now apply sine rule in triangle ABI

$$\frac{AB}{\sin\left(\pi - \left(\frac{B}{2} + \frac{A}{2}\right)\right)} = \frac{BI}{\sin \frac{A}{2}} \Rightarrow \frac{AB}{\cos \frac{C}{2}} = \frac{BI}{\sin \frac{A}{2}}$$

$$\Rightarrow BI = AB \frac{\sin \frac{A}{2}}{\cos \frac{C}{2}}$$

$$\text{Now } \frac{AP}{BI} = \frac{\sin \frac{B}{2} \cos \frac{C}{2}}{\sin \frac{A}{2}}$$

$$\text{Similarly } \frac{AQ}{CI} = \frac{\sin \frac{C}{2} \cos \frac{B}{2}}{\sin \frac{A}{2}}$$

$$\Rightarrow \frac{AP}{BI} + \frac{AQ}{CI} = \frac{\sin\left(\frac{B+C}{2}\right)}{\sin \frac{A}{2}} = \cot \frac{A}{2}$$

19. Ans. (C)

$$\angle BPA = 90^\circ + \frac{B}{3} \quad \angle ABP = \frac{2B}{3}$$

In triangle APB sine law

$$\Rightarrow \frac{AP}{\sin \frac{2B}{3}} = \frac{C}{\sin\left(90 + \frac{B}{3}\right)} = \frac{C}{\sin \frac{B}{3}}$$

$$\Rightarrow AP = \frac{C \sin \frac{2B}{3}}{\sin \frac{B}{3}} = \frac{2C \sin \frac{B}{3} \sin \frac{B}{3}}{\cos \frac{B}{3}}$$

$$AP = 2C \sin \frac{B}{3}$$

20. Ans. (B)

$$T_{(n)} = \frac{n \cdot \frac{(n+1)}{2}}{1^3 + 2^3 + \dots + n^3} = \frac{n(n+1)/4}{n^2(n+1)^2/4} = \frac{1}{n(n+1)}$$

$$\Rightarrow S_{(n)} = \sum T_{(n)} = \sum \left(\frac{1}{n} - \frac{1}{n+1} \right) = \frac{n}{(n+1)}$$

$$\Rightarrow S_{(29)} - S_{(9)} = \frac{29}{30} - \frac{9}{10} = \frac{1}{15}$$

SECTION-II

1. Ans. 3.00

$$\therefore 2 \cos \theta - 1 \geq 0$$

$$\Rightarrow \cos \theta \geq \frac{1}{2} \dots (1)$$

$$\text{Now, } 2 [2 \cos \theta - 1] + 2 \cos^2 \theta - 1 \geq 3$$

$$\Rightarrow 2 \cos^2 \theta + 4 \cos \theta - 6 \geq 0$$

$$\Rightarrow (\cos \theta - 1) (\cos \theta + 3) \geq 0$$

$$\Rightarrow \cos \theta \geq 1 \text{ or } \cos \theta \leq -3$$

$$\Rightarrow \cos \theta = 1 \Rightarrow \theta = 2n\pi, n \in \mathbb{I}$$

$$\Rightarrow \theta = 0, 2\pi, -2\pi (\because \theta \in [-2\pi, 2\pi])$$

2. Ans. 20.00

$$\frac{t_3}{t_6} = \frac{1}{8}$$

$$\frac{ar^2}{ar^5} = \frac{1}{8}$$

$$r^3 = 8$$

$$r = 2$$

$$t_1 + t_5 = 170$$

$$a + ar^4 = 170$$

$$a + 16a = 170$$

$$17a = 170$$

$$a = 10$$

$$t_2 = ar$$

$$= 10 \times 2$$

$$t_2 = 20$$

3. Ans. -4.00

$$R = \log_2 \sin\left(\frac{\pi}{5}\right) + \log_2 \sin\left(\frac{2\pi}{5}\right) + \log_2$$

$$\sin\left(\frac{3\pi}{5}\right) + \log_2 \sin\left(\frac{4\pi}{5}\right) - \log_2 5$$

$$R = \log_2 \left(\sin \frac{\pi}{5} \cdot \sin \frac{2\pi}{5} \cdot \sin \frac{3\pi}{5} \cdot \sin \frac{4\pi}{5} \right) - \log_2 5$$

$$= \log_2 \left(\sin^2 \frac{\pi}{5} \cdot \sin^2 \frac{2\pi}{5} \right) - \log_2 5$$

$$= \log_2 \frac{1}{4} \left(2 \sin^2 \frac{\pi}{5} \cdot 2 \sin^2 \frac{2\pi}{5} \right) - \log_2 5$$

$$= \log_2 \frac{1}{4} \{ (1 - \cos 72^\circ) (1 - \cos 144^\circ) \} - \log_2 5$$

$$= \log_2 \frac{1}{4} \{ (1 - \sin 18^\circ) (1 + \cos 36^\circ) \} - \log_2 5$$

$$= \log_2 \frac{1}{4} \left\{ \left(1 - \frac{\sqrt{5}-1}{4} \right) \left(1 + \frac{\sqrt{5}+1}{4} \right) \right\} - \log_2 5$$

$$= \log_2 \frac{1}{4} \left(\frac{5-\sqrt{5}}{4} \right) \left(\frac{5+\sqrt{5}}{4} \right) - \log_2 5$$

$$= \log_2 \frac{1}{4} \left(\frac{20}{4 \times 4} \right) - \log_2 5 = \log_2 \left(\frac{5}{16} \right) - \log_2 5$$

$$= \log_2 5 - \log_2 16 - \log_2 5 = -4.$$

4. Ans. 8.00

$$\text{Area of square} = S + A + S = 16$$

$$\Rightarrow 2S + A = 16 \dots (i)$$

and area of quarter circle

$$\Rightarrow A + S = \frac{1}{4} \pi (4)^2$$

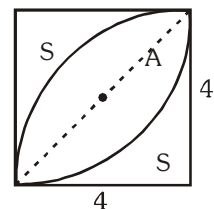
$$\Rightarrow A + S = 4\pi \dots (ii)$$

from (i) and (ii)

$$S = 16 - 4\pi$$

$$\text{and } A = 8\pi - 16$$

$$\Rightarrow \frac{A+16}{\pi} = 8$$



Alternate :

Required area

$$= 2 \{ \text{area of sector AECB} - \text{area of triangle ACB} \}$$

$$= 2 \cdot \left\{ \frac{1}{4} \pi (4)^2 - \frac{1}{2} \cdot 4 \cdot 4 \right\} = 8\pi - 16$$

5. Ans. -1.00

$$4\sin^2\theta + 4\cos^3\theta - 12\sin\theta \cos\theta + 4 = 0$$

$$\sin^3\theta + \cos^3\theta + 1 = 3(\sin\theta \cos\theta + 1)$$

$$\text{possible when } \sin\theta + \cos\theta + 1 = 0$$

6. Ans. 108.00

$$\text{Let } A = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

$$= a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{21}a_{32}a_{13} - (a_{13}a_{22}a_{31} + a_{11}a_{23}a_{32} + a_{12}a_{21}a_{33})$$

$$= x - y \quad (\text{let})$$

$$\text{Now } |A| \text{ is maximum, when } y = 0.$$

$$\Rightarrow \text{one possibility is to choose } a_{21} = a_{32} = a_{13} = 0$$

$$\& \text{ remaining elements to be 3.}$$

$$\therefore |A|_{\max} = 54$$

$$\text{Similarly } |A|_{\min} = -54$$

$$\text{Hence } |A|_{\max} - |A|_{\min} = 54 - (-54) = 108$$

7. Ans. 12.00

$$AP \cdot AQ = AY^2 \Rightarrow AY = 6$$

$$\text{Similarly, } BQ \cdot BP = BR \cdot BS$$

$$\Rightarrow BR(5 + BR) = 3.8 \text{ or } BR^2 + 5BR - 24 = 0$$

$$\Rightarrow (BR + 8)(BR - 3) = 0$$

$$\therefore BR = 3$$

$$\therefore AB = 12, BC \text{ must be 9 units.}$$

$$\text{Hence } CS = 1$$

$$\therefore CZ^2 = CS \cdot CR = 1.6$$

$$\Rightarrow AY + CZ^2 = 12$$

8. Ans. 10.00

$$(\alpha - 2)^2 + (\beta - 3)^2 = 5^2$$

$$\Rightarrow (\alpha, \beta) \text{ lies on } (x - 2)^2 + (y - 3)^2 = 5^2$$

$$\text{Let } \alpha = 2 + 5\cos\theta$$

$$\beta = 3 + 5\sin\theta$$

$$12\alpha + 16\beta = 12(2 + 5\cos\theta) + 16(3 + 5\sin\theta)$$

$$= 4\{15\cos\theta + 20\sin\theta + 18\}$$

$$= 4\{5(3\cos\theta + 4\sin\theta) + 18\}$$

$$(12\alpha + 16\beta)_{\max} = 4\{5.5 + 18\} = 172$$

9. Ans. 3.00

$$\sin^6\theta + \cos^6\theta = \frac{1}{2} \Rightarrow 1 - 3\sin^2\theta \cos^2\theta = \frac{1}{2}$$

$$\Rightarrow \sin^2\theta \cos^2\theta = \frac{1}{6}$$

$$\sin^4\theta + \cos^4\theta = \lambda_2$$

$$\Rightarrow \lambda_2 = 1 - 2\sin^2\theta \cos^2\theta = \frac{2}{3}$$

$$\lambda_3 = \sin^4\theta \cos^4\theta = \frac{1}{36}$$

$$\frac{\lambda_1}{10} = \sin^6\theta \cos^6\theta = \frac{1}{216} \Rightarrow \lambda_1 = \frac{5}{108}$$

$$\Rightarrow 36\lambda_1 + \lambda_2 + 24\lambda_3 = \frac{5}{3} + \frac{2}{3} + \frac{2}{3} = 3$$

10. Ans. 589.00

$$S_{20} = \left\{ \underbrace{\frac{399}{20}, \frac{419}{20}, \frac{439}{20}, \dots}_{20 \text{ terms}} \right\}$$

$$\Rightarrow \text{Sum of the elements of}$$

$$S_{20} = \frac{20}{2} \left[2 \cdot \frac{399}{20} + 19.1 \right] = 589$$